KSHITIZ RAY

MECHATRONICS ENGINEERING STUDENT | PLC CONTROLS | ROBOTICS | MANUFACTURING

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LinkedIn | GitHub | Portfolio

SUMMARY

Mechatronics engineering student at KIIT University with internship exposure to industrial robotics, manufacturing automation and material handling. Supported FANUC robot setup, testing and commissioning, and contributed to a safety-focused tanker-unloading automation concept at Tata Steel. Confident in PLC programming with Siemens TIA Portal, with additional embedded-systems and prototyping projects; seeking a Cummins engineering internship in controls, automation, manufacturing or related mechatronics roles.

TECHNICAL SKILLS

PLC & Controls: PLC programming, Siemens TIA Portal, control systems, industrial sensing
Robotics & Automation: FANUC ARC Mate, FANUC LR Mate, Elite Cobots, robot setup and commissioning, robotic welding and spray-paint automation
Mechanical & Manufacturing: SolidWorks, Siemens NX, Fusion 360, Mastercam, mechanical assembly, MIG welding, 3D printing, Ultimaker Cura
Embedded & IoT: Embedded C, Arduino, ESP32, Raspberry Pi, Arduino IDE, Arduino IoT Cloud, Tinkercad, sensor-based monitoring

EXPERIENCE

Engineering Intern | Tata Steel | May–June 2026

- Contributed to a safety-focused automation concept for unloading desulphurization (DS) material from tanker trucks, targeting reduced manual hose handling and safer operator workflows.
- Explored how robotic material handling and industrial sensing could support the unloading process.

Junior Project Engineer Intern | COTMAV Robotics & Automation | September 2024–March 2025

- Supported installation, mechanical setup, testing and commissioning of FANUC ARC Mate and FANUC LR Mate industrial robots in manufacturing environments.
- Contributed to robotic welding and spray-painting automation projects involving Endurance, Studds Helmets, Highco Engineering Works, Arvind Mill and RKFL.

SELECTED PROJECTS

- **4-DOF SCARA Robotic Arm** — Designed the joint and actuator layout for repeatable motion; worked with NEMA 17 stepper motors, GRBL and Arduino IDE.
- **Delta 3D Printer & Smart Enclosure** — Designed and assembled a delta printer, tuned Ultimaker Cura settings, and added relay-based remote control for enclosure power, lighting and camera.
- **Wi-Fi Smart-Home Automation** — Built a Wi-Fi-enabled household-appliance control system with a mobile interface; explored smart-assistant integration.
- **RoboDog (Four-Legged Quadruped)** — Developing an ESP32-powered robot with Wi-Fi control, IMU-based stabilization, autonomous navigation and 3D-printable parts.

EDUCATION

B.Tech, Mechatronics Engineering | KIIT University, Bhubaneswar | 2025–Present

Diploma, Mechatronics Engineering | NTTF R.D. Tata Technical Education Centre
Jamshedpur | 2021–2024 | CGPA: 7.82/10

CERTIFICATIONS & RECOGNITION

Certifications

- NOCN Level 5 Diploma in Mechanical Engineering — Mechatronics & Smart Manufacturing (Merit)
- NSDC Technician Instrumentation — NSQF Level 4
- NSDC Wireman, Control Panel Electronics — NSQF Level 3
- Autodesk Fusion 360 Cloud — Prelims participation
- Student AI Masterclass — upEducators / Google for Education program (2026)
- Digi-Shala: Measuring Instruments — Tata Steel Technical Institute (2026)
- Digi-Shala: Engineering Drawing — Tata Steel Technical Institute (2026)
- Digi-Shala: Limit, Fit & Tolerance — Tata Steel Technical Institute (2026)
- Digi-Shala: Bearings — Tata Steel Technical Institute (2026)

Awards & Recognition

- Finalist, Autodesk Skill Surge Inter-State Design Challenge (2025)
- 1st Rank, Youth Skill Competition
- Technical Project Model Display, Tata Steel Technical Institute
- NTTF Home Automation Achievement